ID	Requirement	Priority	Actor(s)	Acceptance Criteria
FR-01	The system shall allow users to register by providing required personal information.	High	User/Donor	Valid submission creates an account with a unique user ID.
FR-02	The system shall allow registered users to log in using valid credentials.	High	All Users	Valid credentials grant access; invalid ones are rejected with a clear error.
FR-03	The system shall allow authorized staff to register donors with donor ID, name, blood group, contact,	High	Blood Bank Staff	Donor record is created with all required fields and a unique donor ID.
FR-04	The system shall allow authorized staff to add, update, search, and view donor records.	High	Blood Bank Staff	All four operations succeed per role permissions.
FR-05	The system shall allow staff to record blood donations with donation ID, donor ID, blood group,	High	Blood Bank Staff	Donation record is created with all required fields and a unique donation ID.
FR-06	The system shall allow administrators to add, update, and deactivate blood banks with name,	High	Administrator	Admin can create, modify, and deactivate blood-bank records.
FR-07	The system shall allow authorized staff to manage blood inventory by blood group and blood bank.	High	Blood Bank Staff	Staff can view and update inventory records per role permissions.
FR-08	The system shall automatically update inventory quantity on every donation and issue transaction.	High	System	Inventory reflects the correct quantity after each valid transaction.
FR-09	The system shall allow users to search for available blood by blood group, location, and required	High	All Users	Search returns matching blood banks within performance target.
FR-10	The system shall allow users to submit blood requests with blood group, quantity, location, urgency,	High	User/Requester	Valid request is saved and assigned a unique request ID.
FR-11	The system shall allow authorized staff to view and update blood-request statuses (pending,	High	Blood Bank Staff	Staff can view and change status per role permissions.
FR-12	The system shall alert authorized staff when a blood group falls below its configured minimum	Medium	System/Staff	Alert is generated as soon as inventory crosses threshold.
FR-13	The system shall provide notifications for events including new requests, request-status	Medium	System/All Users	Relevant users receive the correct notification on each configured event.
FR-14	The system shall allow authorized users to generate reports on donors, donations, inventory,	Medium	Staff/Admin	Authorized users can generate and view the specified reports.
FR-15	The system shall allow authorized users to search and filter records by donor ID, donation ID, blood group,	Medium	Staff/Admin	Filter returns records matching all selected criteria.
FR-16	The system shall enforce role-based permissions for donors, staff, and administrators.	High	System/Admin	Each role accesses only permitted functions.